

1. Introduction

This project demonstrates the application of regression and classification techniques on real-world datasets using Python libraries such as `scikit-learn`, `pandas`, `matplotlib`, and `seaborn`. The assignment involves building and evaluating:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Logistic Regression (for classification)

2. Regression Analysis

2.1 Dataset Description

- **Dataset Name:** Car Prices Dataset
- **Source:** Kaggle
- **Target Variable:** Car Price
- **Features Used:** [e.g., Engine Size, Horsepower, Mileage]

2.2 Exploratory Data Analysis (EDA)

- Data cleaning: Handling missing values, data type conversions.
- Visualizations: Histograms, scatter plots, correlation heatmaps.
- Insights gained from the EDA.

2.3 Preprocessing

- Feature selection and scaling.
- Splitting the dataset into training and testing sets (e.g., 80/20).

2.4 Simple Linear Regression

- Chosen feature: [e.g., Horsepower]
- Regression line plot.

- Metrics: R², MSE

2.5 Multiple Linear Regression

- Chosen feature: [e.g., Horsepower]
- Regression line plot.
- Metrics: R², MSE
- Interpretation of results.

2.6 Polynomial Regression

- Applied on the best-performing feature.
- Polynomial degrees: 2, 3, and 4.
- Metrics and visual comparison of performance.
- Insights on overfitting or underfitting.

3. Classification Using Logistic Regression

3.1 Dataset Description

- **Dataset Name:** Titanic
- **Source:** Kaggle
- **Target Variable:** Survived (Yes/No)

3.2 Preprocessing

- Encoding categorical features.
- Scaling if needed.
- Splitting into training and testing sets.

3.3 Logistic Regression Model

- Training the model.
- Making predictions on the test set.

3.4 Model Evaluation

- Confusion matrix (visualized using seaborn/matplotlib).
- Accuracy, Precision, Recall, F1 Score.
- Discussion on model performance.

4. Conclusion

- Summary of key findings from regression and classification models.
- Which model performed best and why?
- Challenges encountered and how they were resolved.
- Possible improvements or next steps (e.g., try other ML algorithms).

Team Members

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